Homework #0 — Introduction and Checklist (25 pts)

MATH 241 — Probability, Fall 2026

Due Date: Friday, 9/4 at 11:59pm ET

Instructions

- Please submit this assignment to Gradescope as a .pdf format. Any submissions not in this format are subjected to point deductions.
- Responses can be hand-written or typed.
- Certain questions will ask you to do the problem in a specific way. Please adhere to these instructions or points will be deducted.
- For all questions involving manual calculations, please show all work and round answers to four decimal places.
- Refer to syllabus for late work deductions.

Background

The zeroth homework is an introductory assignment that allows you to get familiar with all technological components of this course and allows me to learn a little about you. Additionally, there is a quick probability problem to gear you up for the course.

Part I: Technology (10 pts)

Complete the following things in this checklist:

1. Install and download R and RStudio. Insert a screenshot of your RStudio screen.
   - Instructions
2. Register and login to Slack (for the correct section!).
   - Instructions are in the syllabus! Under the *random* channel, answer the following question I've posted.
3. Read the syllabus on my website: Link

- Provide one piece of information that is useful to you as we go throughout the semester.

Part II: Survey (10 pts)

Fill out this Google Form, which contains a brief survey about your skills and expectations going into this course, as well as some brief questions about yourself.

1. Survey

Part III: Introduction into Probability (5 pts)

It's 6pm and you are about to go on a run. You notice that there are ominous clouds up in the sky. You're unsure if you want to run outside if you know it might rain. In that moment, how could I probabilistically determine whether I should run outside? Also consider:

- What are some variables/characteristics I would need to consider?
- What has history told you before?
- Can I truly be certain that it will rain or not?

Answer the following question (and considerations) in two ways:

1. Sketch out a response to this question, using your own intuition.
2. Prompt a large-language model (or a person) and put their response below.